

Module5_Lab1

October 1, 2026

```
[1]: import pandas as pd
df = pd.read_csv('chickwts.csv')

df.groupby("feed")["weight"].mean().sort_values()
```

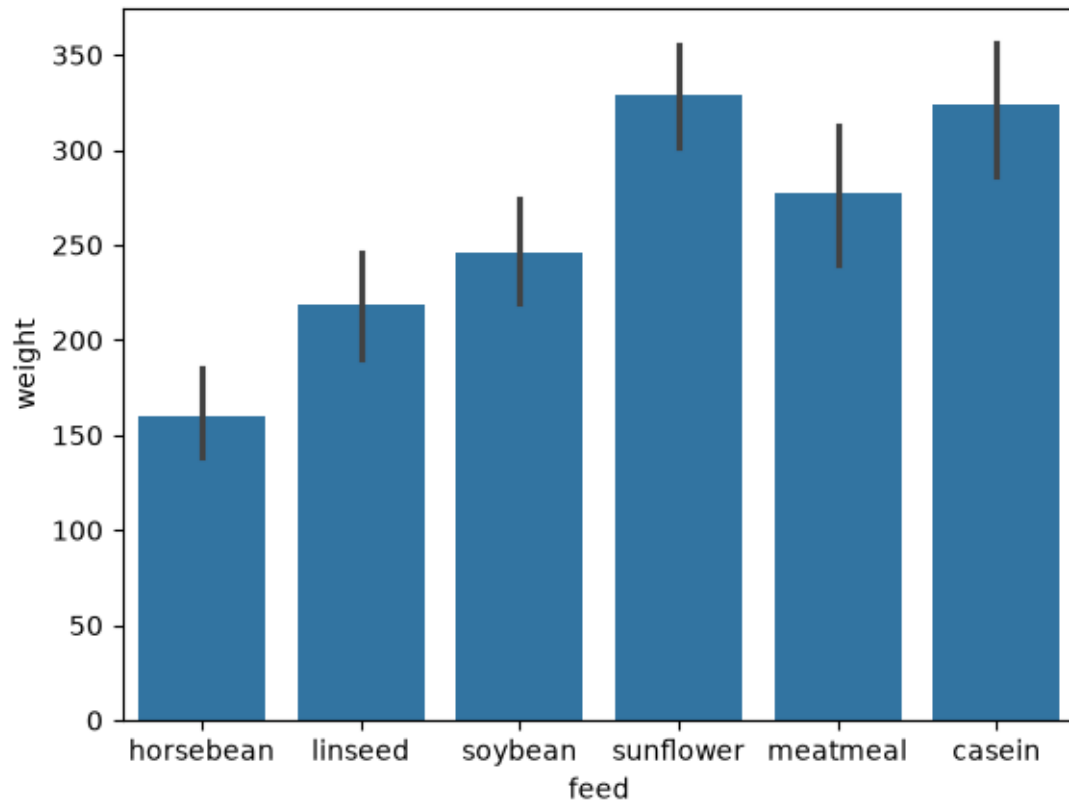
```
[1]: feed
horsebean    160.200000
linseed      218.750000
soybean      246.428571
meatmeal     276.909091
casein       323.583333
sunflower    328.916667
Name: weight, dtype: float64
```

```
[2]: import seaborn as sns
import matplotlib.pyplot as plt
sns.barplot(data=df, x="feed", y="weight")
plt.show()
```

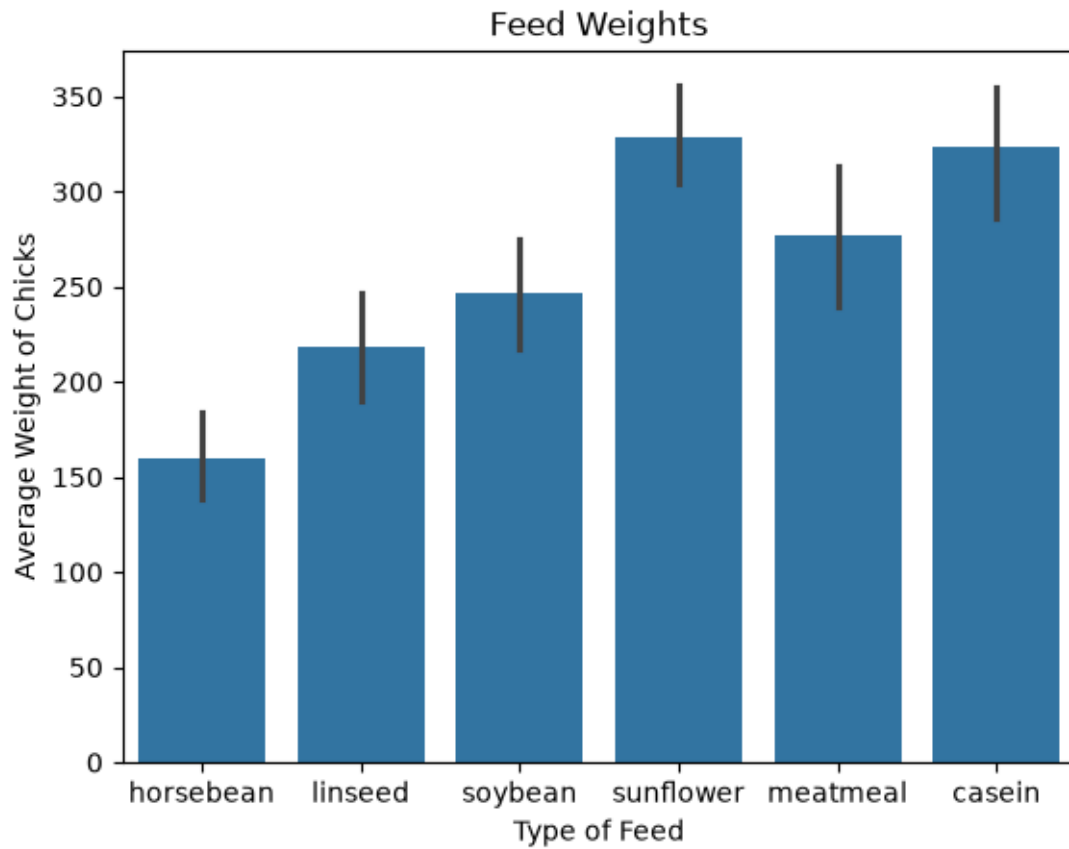
```
Cell In[2], line 2
import matplotlib.pyplot as plt
~
```

SyntaxError: invalid syntax

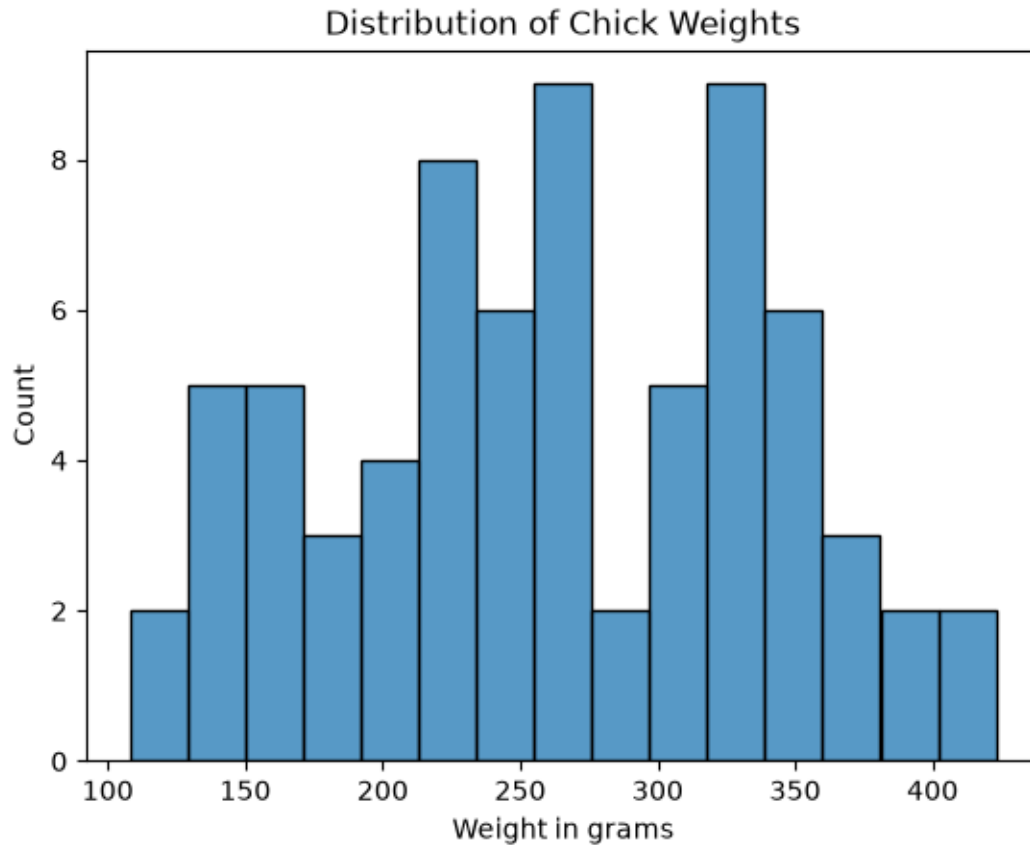
```
[3]: import seaborn as sns
import matplotlib.pyplot as plt
sns.barplot(data=df, x="feed", y="weight")
plt.show()
```



```
[4]: sns.barplot(data=df, x="feed", y="weight")
plt.title("Feed Weights")
plt.xlabel("Type of Feed")
plt.ylabel("Average Weight of Chicks")
plt.show()
```



```
[5]: sns.histplot(data=df, x="weight", bins=15)
plt.title("Distribution of Chick Weights")
plt.xlabel("Weight in grams")
plt.ylabel("Count")
plt.show()
```



```
[6]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df['Date'] = pd.to_datetime(df['Date'])
df_melted = df.melt(id_vars=['Date'], var_name='Museum', value_name='Visitors')

sns.lineplot(data=df_melted, x='Date', y='Visitors')

plt.title('Monthly Museum Visitors Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Visitors')
plt.tight_layout()
plt.show()
```

```
-----
KeyError                                Traceback (most recent call last)
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3641, in Index.
    get_loc(self, key)
    3640 try:
```

```
-> 3641     return self._engine.get_loc(casted_key)
    3642 except KeyError as err:
```

```
File pandas/_libs/index.pyx:168, in pandas._libs.index.IndexEngine.get_loc()
--> 168 'Could not get source, probably due dynamically evaluated source code.'
```

```
File pandas/_libs/index.pyx:197, in pandas._libs.index.IndexEngine.get_loc()
--> 197 'Could not get source, probably due dynamically evaluated source code.'
```

```
File pandas/_libs/hashtable_class_helper.pxi:7668, in pandas._libs.hashtable.
↳PyObjectHashTable.get_item()
-> 7668 'Could not get source, probably due dynamically evaluated source code.'
```

```
File pandas/_libs/hashtable_class_helper.pxi:7676, in pandas._libs.hashtable.
↳PyObjectHashTable.get_item()
-> 7676 'Could not get source, probably due dynamically evaluated source code.'
```

KeyError: 'Date'

The above exception was the direct cause of the following exception:

KeyError Traceback (most recent call last)

Cell In[6], line 5

```
1 import pandas as pd
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4
----> 5 df['Date'] = pd.to_datetime(df['Date'])
      6 df_melted = df.melt(id_vars=['Date'], var_name='Museum',
↳value_name='Visitors')
      7
      8 sns.lineplot(data=df_melted, x='Date', y='Visitors')
```

```
File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:4378, in DataFrame.
```

```
↳__getitem__(self, key)
4374
4375     if is_single_key:
4376         if self.columns.nlevels > 1:
4377             return self._getitem_multilevel(key)
-> 4378         indexer = self.columns.get_loc(key)
4379         if is_integer(indexer):
4380             indexer = [indexer]
4381     else:
```

```
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3648, in Index.
```

```
↳get_loc(self, key)
3643     if isinstance(casted_key, slice) or (
3644         isinstance(casted_key, abc.Iterable)
```

```

3645         and any(isinstance(x, slice) for x in casted_key)
3646     ):
3647         raise InvalidIndexError(key) from err
-> 3648     raise KeyError(key) from err
3649 except TypeError:
3650     # If we have a listlike key, _check_indexing_error will raise
3651     # InvalidIndexError. Otherwise we fall through and re-raise
3652     # the TypeError.
3653     self._check_indexing_error(key)

```

KeyError: 'Date'

```

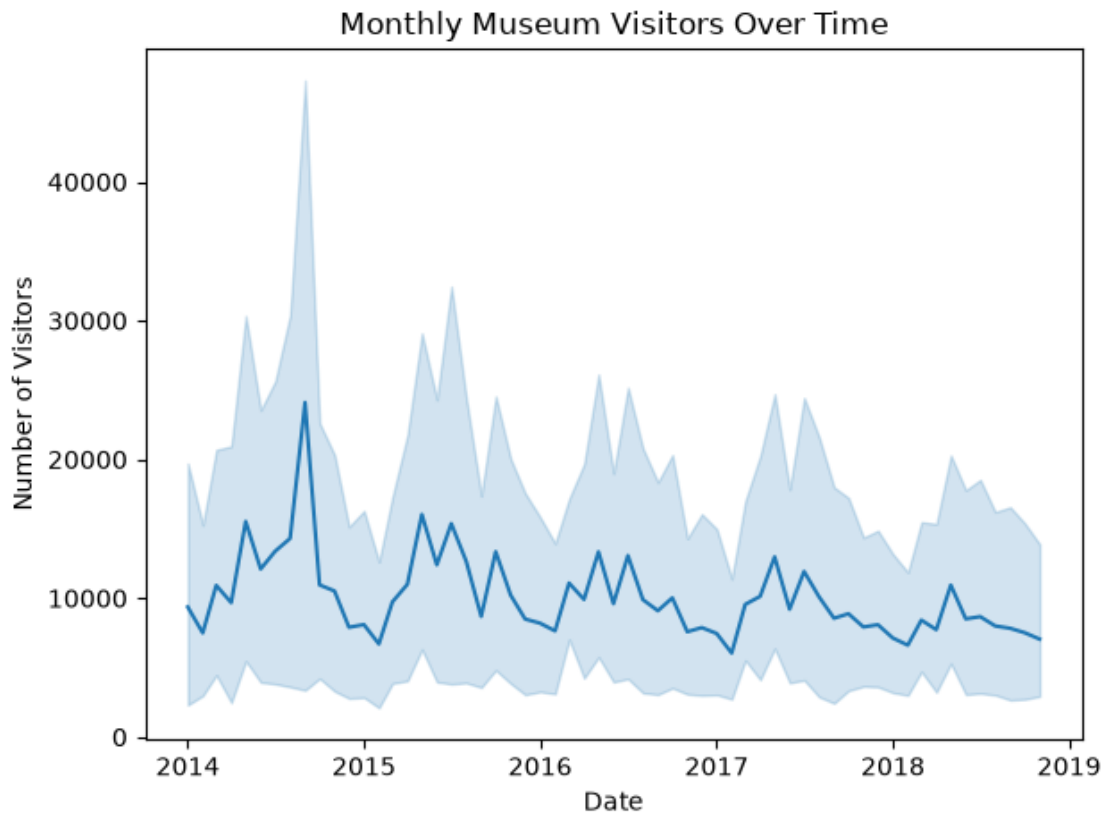
[7]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv('museum_visitors.csv')

df['Date'] = pd.to_datetime(df['Date'])
df_melted = df.melt(id_vars=['Date'], var_name='Museum', value_name='Visitors')

sns.lineplot(data=df_melted, x='Date', y='Visitors')

plt.title('Monthly Museum Visitors Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Visitors')
plt.tight_layout()
plt.show()

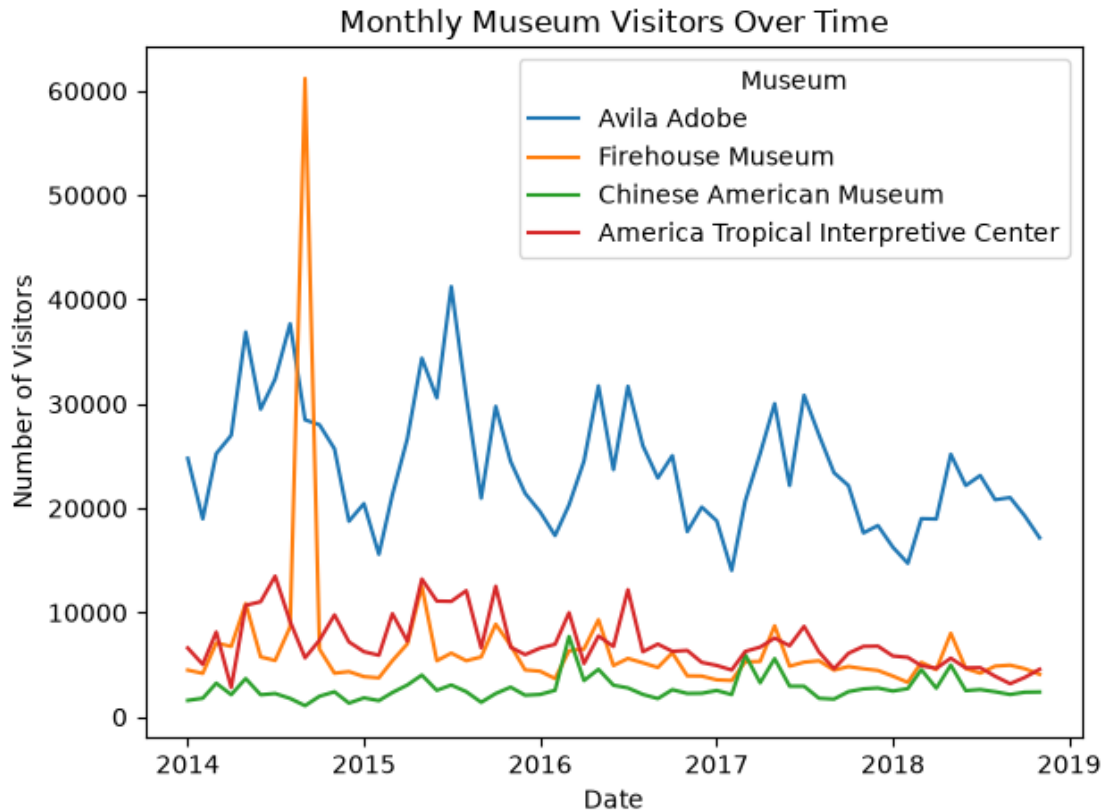
```



```
[8]: df['Date'] = pd.to_datetime(df['Date'])
df_melted = df.melt(id_vars=['Date'], var_name='Museum', value_name='Visitors')

sns.lineplot(data=df_melted, x='Date', y='Visitors', hue='Museum')

plt.title('Monthly Museum Visitors Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Visitors')
plt.tight_layout()
plt.show()
```

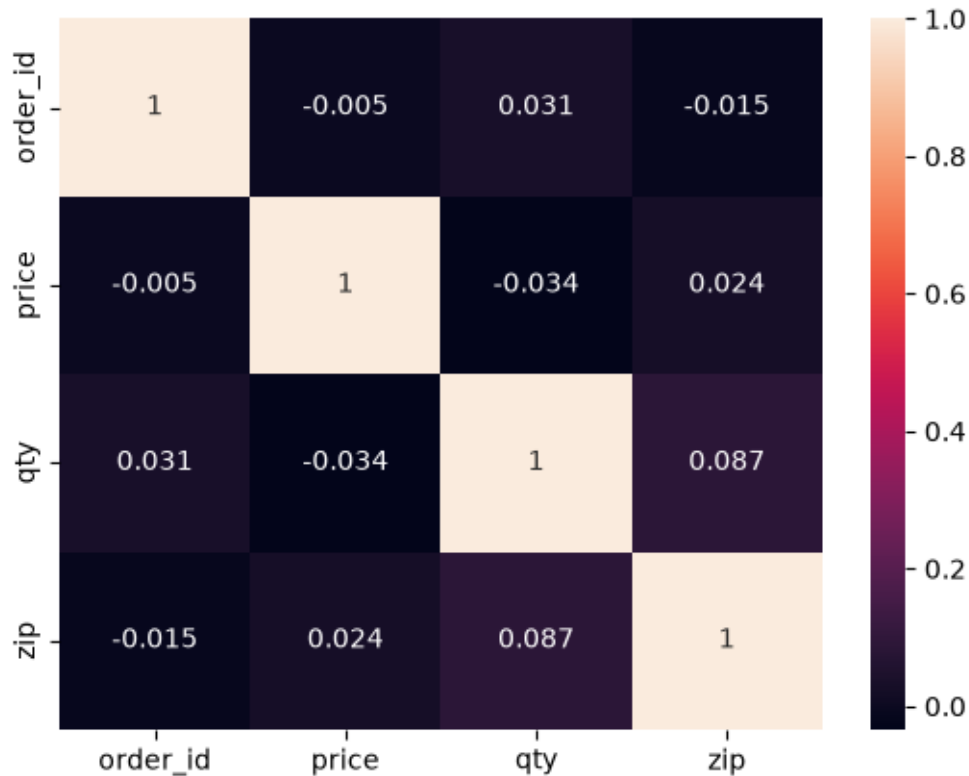


Two things stood out to me: there seems to be a huge spike in number of visitors for the Firehouse Museum between 2014 and 2015. Another thing is that it appears that there is a downward trend in number of visitors over the years for all museums.

```
[9]: import pandas as pd
import seaborn as sns
df = pd.read_csv("sales_clean.csv")

sns.heatmap(df.corr(numeric_only=True), annot=True)
```

[9]: <Axes: >



Hypothesis: there is no difference in the mean weight of sunflower-fed chicks vs horsebean-fed chicks.

```
[10]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('chickwts.csv')

from scipy.stats import ttest_ind
a = df[df["feed"] == "sunflower"]["weight"]
b = df[df["feed"] == "horsebean"]["weight"]
ttest_ind(a, b)
```

```
[10]: TtestResult(statistic=np.float64(8.848346742516636),
pvalue=np.float64(2.3743507676837164e-08), df=np.float64(20.0))
```

The t is 8.85, p-value is 2.37×10^{-8} . The t is indeed over 7, which means that there is a significant difference between the two feeds, and also the sunflower feed weighed more than the horsebean feed, as the p-value indicated (way less than 0.05). Perhaps there might not be enough data to generalize?